LAB 6: Develop Software for Basic Robot Navigation with a Finite State Machine

Introduction
In this lab, you will develop software from a finite state machine representation.

Note, you are not required to use the structured FSM implementation from lecture, but you must implement the code based on your FSM.

Specifications
Your finite state machine has 7 states.

StartUp: Wait for SW1 or SW2 to be pressed. If either pressed, transition to Forward
Forward: Drive forward until a bump sensor is pressed. When a bump sensor is pressed, transition to WhichButtons
TurnLeft: Turn left approximately 90 degrees. Transition to Forward.
TurnRight: Turn right approximately 90 degrees. Transition to Forward.
WhichButtons: Identify which buttons are pressed. If the buttons pressed are on the left side of the robot, transition to TurnRight. If the buttons pressed on the right side of the robot, transition to TurnLeft. If there are buttons left and right, transition to Backward.
Backward: Back up approximately 6 inches. Transition to Turn180
Turn180: Turn approximately 180 degrees. Transition to Forward.

With your lab team, draw the FSM state transition graph.

Write code which will correctly set up the robot for PWM motor control, as well as initialize the bumpers. The bump switches pin out can be found in the ti-rslk-max-chassis-board-v1.0-schematic.pdf in the Datasheets module/link. Then write code to implement the FSM described above.

> ***Checkpoint 0:*** *Show the TA or the instructor your FSM code. Save the screenshots in your lab notebook.*

Test your robot to validate the behavior. Test and record each behavior. Use the debugger on Code Composer Studio to validate that state transitions operate correctly. Record data/screenshots showing this.

Then, take your robot to the playfield and test. Again, verify operation and state transitions for all states. Conduct each test at least 3 times. Record data for validation.

> ***Checkpoint 1:*** *Show the TA or instructor your working robot, able to navigate without getting stuck for at least 90 seconds.*

Generate a report, including your FSM, code, and testing/validation procedures and results. This report will be used to evaluate R8*. Submissions will be individual.

R8* Rubric

Requirement	Unacceptable	Minimal	Acceptable	Excellent	Comments
Diagram of FSM state transition graph					
Code complete, accurate, and commented					
Validation testing using debugger documented					
Validation testing documented (3 times, all states)					
Changes made as a result of validation testing (if applicable- please state)					
Check sheet completed (performance)					